

Soil Health & Crop Recommendation Report

Report Date: November 07, 2025 at 18:17:34 UTC

Farmer Information	
Name:	rajubhosla
Phone:	8088117817
Location:	bijapur
Land Size:	23.00 acres

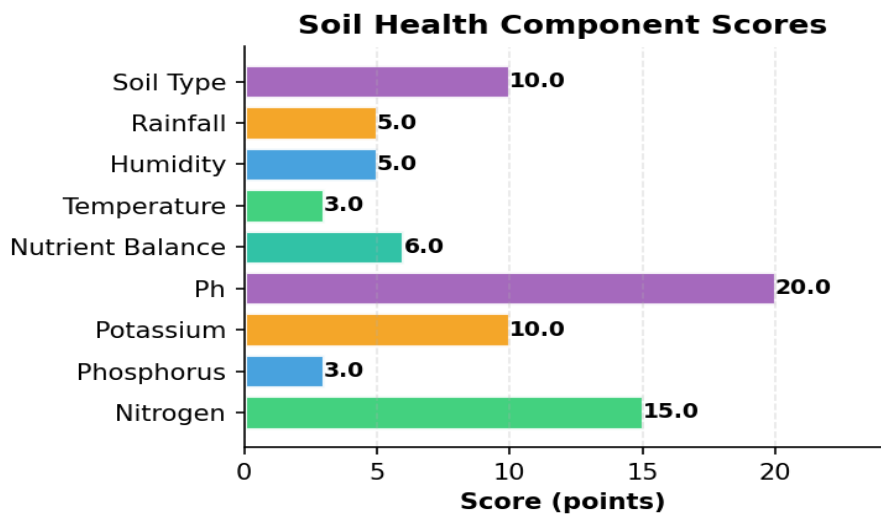
Parameter	Value	Unit
Nitrogen (N)	88.0	mg/kg
Phosphorus (P)	96.0	mg/kg
Potassium (K)	96.0	mg/kg
pH Level	7.30	
Temperature	32.6	°C
Humidity	73.0	%
Rainfall	240.0	mm
Soil Type	Alluvial Soil	

Soil Health Assessment

Overall Soil Health Score		77.0/100	Good
Component	Score	Points	
Nitrogen	15.0	points	
Phosphorus	3.0	points	
Potassium	10.0	points	
Ph	20.0	points	
Nutrient Balance	6.0	points	
Temperature	3.0	points	
Humidity	5.0	points	
Rainfall	5.0	points	
Soil Type	10.0	points	

Soil Health & Crop Recommendation System

Comprehensive Agricultural Analysis Report



Recommended Crop

Predicted Crop	Rice
Confidence Level	40.1%

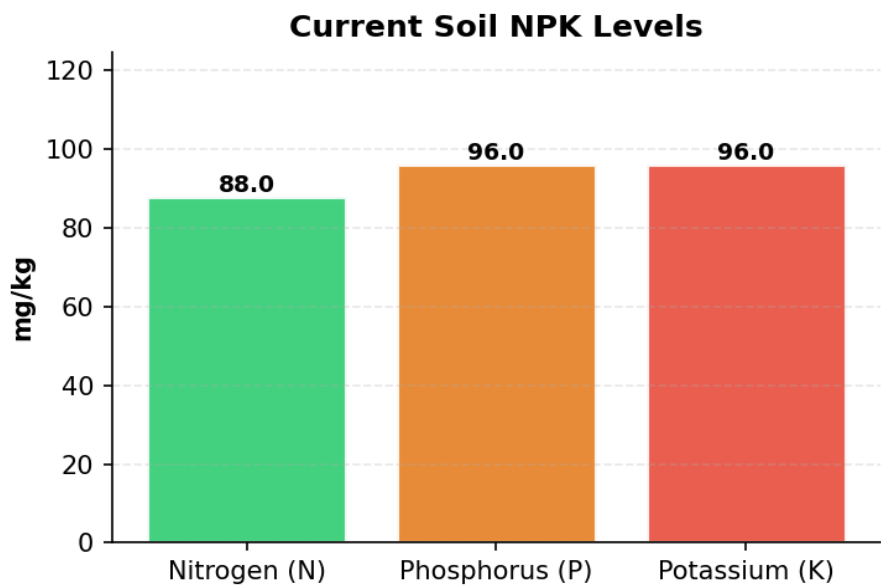
Alternative Crop Options

Rank	Crop	Confidence (%)
1	Banana	18.9%
2	Jute	12.3%

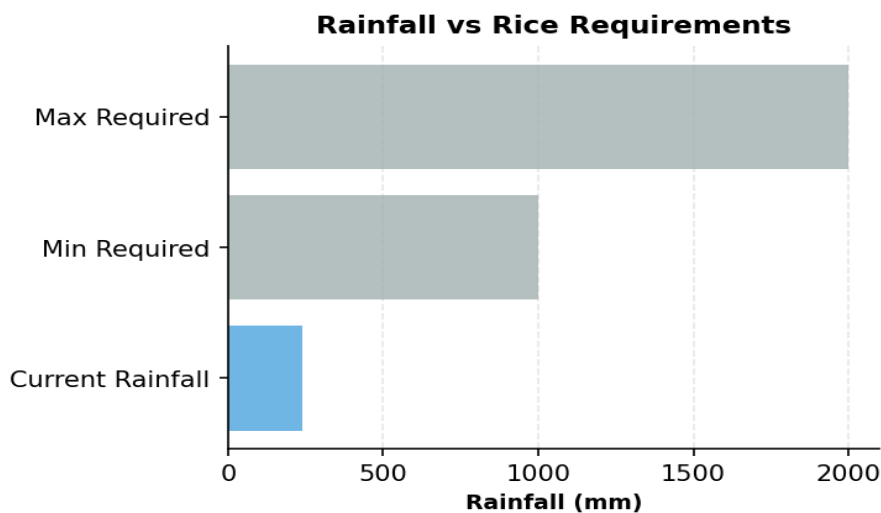
Soil NPK Levels Visualization

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Rainfall Analysis



Important Warnings

- Very high phosphorus levels detected. May interfere with micronutrient uptake.

Soil Treatment Recommendations

- Soil pH (7.3) is above optimal range (5.5-7.0) for rice. Add gypsum (1-2 tons/acre) or organic compost to lower pH.
- Nitrogen (88.0 mg/kg) is within optimal range for rice.
- Phosphorus (96.0 mg/kg) is excessive for rice. Optimal range: 30-50. High P can interfere with micronutrient uptake.

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- Potassium (96.0 mg/kg) is excessive for rice. Optimal range: 30-50. High K can cause magnesium deficiency.
- Rich in nutrients, good water retention. Suitable for most crops.

Fertilizer Recommendations

Fertilizer Requirements for rice:

- Land Size: 23.0 acres
- Current NPK: N=88.0, P=96.0, K=96.0 mg/kg
- Ideal NPK: N=90, P=40, K=40 kg/acre

Required Application:

- Nitrogen (N): 46.0 kg total (2.0 kg/acre)
→ Apply in 2-3 split doses: 50% at sowing, 25% at tillering, 25% at flowering

Organic Alternatives:

- Compost/FYM: 5-10 tons/acre (provides N, P, K)
- Green Manure: Leguminous crops (fixes N)
- Biofertilizers: Rhizobium (N), PSB (P), VAM (P)

Crop-Specific Tips for Rice

Ensure consistent water supply and transplant in puddled field. Maintain 5-7 cm water depth.

Additional Notes

- These recommendations are based on comprehensive soil analysis and advanced ML predictions.
- Actual results may vary based on local conditions, weather patterns, and farming practices.
- Consult with local agricultural experts before making major farming decisions.
- Regular soil testing (every 2-3 years) is recommended for optimal crop management.
- Consider crop rotation and organic farming practices for long-term soil health.
- Monitor weather forecasts and adjust irrigation schedules accordingly.